

ANKUSH KUNDAPURA ANNAIAH

SDE Intern · RL Environments · AI Systems · Python · Simulation & Tooling

ankushkundapuraannaiah@gmail.com | Bengaluru, India | linkedin.com/in/ankushkundapura-annaiah-1797ba384 | github.com/ankushkundapuraannaiah-bit

PROFESSIONAL SUMMARY

Passionate B.Tech CSE (AI & ML) student and GenAI Developer with hands-on experience building scalable AI pipelines, automation tooling, REST API integrations, and LLM-powered systems in Python. Solid understanding of Reinforcement Learning fundamentals — including MDPs, reward functions, policy design, and value functions — backed by deep learning coursework using PyTorch and TensorFlow. Experienced in designing modular, low-latency execution pipelines, sandboxed environments, and integration SDKs. Eager to apply these skills as an SDE Intern in RL Environments to build simulation infrastructure that enables effective agent training and evaluation at scale.

EXPERIENCE

GenAI Developer Intern · HiDevs · Bengaluru, India

Feb 2026

- ▶ Built low-latency AI execution pipelines and tool automation scripts in Python using LangChain and Hugging Face Transformers — directly applicable to RL tool execution pipeline development.
- ▶ Designed and maintained RAG (Retrieval-Augmented Generation) pipelines with modular, sandboxed components integrating vector databases and LLMs — mirroring sandboxed runtime and environment state management patterns used in RL systems.
- ▶ Developed REST API integrations and internal automation tooling; implemented request/response schemas with structured error handling — aligned with MCP-compatible tool replica development.
- ▶ Built reusable Python helper libraries and automation modules to extend AI tool capabilities and support consistent experimentation across team workflows.
- ▶ Implemented structured logging, experiment tracking, and reproducible evaluation pipelines — analogous to regression runners and scenario replay systems in RL environments.
- ▶ Collaborated directly with AI researchers to understand environment requirements and translate them into working software prototypes, iterating rapidly based on feedback.

Python Developer Intern · Pinnacle Labs · Bengaluru, India

Jan 2026 – Feb 2026

- ▶ Developed robust Python data preprocessing pipelines with modular design, error handling, and transformation logic — foundational for RL environment state management and transition modeling.
- ▶ Built and integrated REST APIs for backend data workflows, demonstrating proficiency in designing scalable service interfaces and integration APIs.
- ▶ Applied structured data engineering practices: schema design, data validation, and pipeline testing — skills directly transferable to environment side-effect modeling and test harness development.

PROJECTS

RL Environment Simulation – Agent Training Sandbox · Python · PyTorch · OpenAI Gym · Custom MDP Design

2026

- ▶ Designed a modular simulation environment implementing a custom MDP (Markov Decision Process) with configurable state spaces, reward functions, transition dynamics, and terminal conditions for RL agent training.
- ▶ Implemented environment state management, side-effect modeling, and episode reset logic following standard Gym-compatible interface patterns — enabling plug-and-play compatibility with RL training loops.
- ▶ Built automated test harnesses and scenario replay systems to validate environment correctness, reward consistency, and edge-case handling across agent policy rollouts.
- ▶ Developed a lightweight internal SDK with helper APIs for environment extension, custom reward shaping, and observation space configuration — enabling rapid experiment iteration.

RAG Pipeline & Sandboxed Tool Execution System · Python · LangChain · Vector DB · REST APIs

2026

- Architected a sandboxed tool execution pipeline for LLM agents — isolating tool calls, managing execution state, and enforcing strict request/response schemas with structured error propagation.
- Built mock service layers and modular tool replicas with consistent API contracts, enabling agent experimentation without live service dependencies — directly analogous to MCP-compatible tool replica design.
- Implemented environment state transitions across multi-turn agent interactions, tracking context, tool outputs, and side-effects across execution steps.

Browser/UI Automation & Interaction Simulation · Python · Selenium · HTML · CSS · Scripting

2025–2026

- Built browser and UI simulation layers supporting programmatic click, type, scroll, and navigation actions — enabling automated interaction testing and agent-driven UI traversal.
- Developed regression test runners and scenario replay scripts to validate UI interaction flows, catching regressions across interface changes with minimal manual effort.
- Designed modular interaction APIs that abstract browser-specific behavior, enabling environment extensions to support new UI components without rewriting core logic.

LLM Fine-Tuning & Evaluation Framework · PyTorch · Hugging Face · Scikit-learn

2026

- Built end-to-end model training, evaluation, and benchmarking pipelines in PyTorch — demonstrating proficiency with deep learning frameworks required for RL policy and value function modeling.
- Implemented automated evaluation datasets, test case generation, and performance regression tracking to ensure consistent model quality across training iterations.

OpenHack 2026 & Codeathon – AI/ML Competitions · Python · Rapid Prototyping · Team Engineering

2026

- Engineered working AI prototypes in competitive hackathon environments — demonstrating problem-solving mindset, rapid tooling development, and collaborative software engineering under pressure.

REINFORCEMENT LEARNING KNOWLEDGE

- Markov Decision Processes (MDPs): state spaces, action spaces, transition functions, discount factors, and episode termination conditions.
- Reward Functions: sparse vs. dense rewards, reward shaping, and goal-conditioned reward design for guiding agent behavior.
- Policy & Value Functions: policy gradient methods, Q-learning, value iteration, and actor-critic architectures.
- RL Frameworks: familiarity with OpenAI Gym / Gymnasium environment interface standards and standard RL training loop design.
- Exploration Strategies: epsilon-greedy, entropy-based exploration, and intrinsic motivation for environment coverage.

TECHNICAL SKILLS

Languages: Python (Advanced) · Bash · SQL · HTML · CSS

RL & AI Systems: MDP Design · Reward Functions · Policy/Value Functions · RL Environment Design · OpenAI Gym · Agent Training Pipelines

Systems & Tooling: Low-Latency Pipeline Design · Sandboxed Runtimes · Mock Services · REST API Integration · MCP-Compatible Tool Design · SDK Development

Testing & Automation: Test Harness Design · Regression Runners · Scenario Replay · Browser/UI Simulation · Automated Benchmarking · Structured Error Handling

AI & ML Frameworks: PyTorch · TensorFlow · Scikit-learn · LangChain · Hugging Face Transformers · NumPy · Pandas

Engineering Practices: Modular System Design · State Management · Side-Effect Modeling · Experiment Tracking · Documentation · Git · Agile

EDUCATION

B.Tech – Computer Science Engineering (AI & ML) · REVA University, BengaluruSep 2025 – Sep 2029

- ▶ Core coursework: Reinforcement Learning, Deep Learning, Algorithm Design, Data Structures, Software Engineering, AI Systems, and Scalable Computing.
- ▶ Active in AI/ML competitions (OpenHack 2026, Codeathon by AiFi) and engineering communities; independently builds and experiments with RL and AI systems.

CERTIFICATIONS & ACHIEVEMENTS

- ▶ Team Essentials for Designing AI Solutions — AI system architecture, tool design, and deployment best practices
- ▶ Data Streaming Engineer Foundations — real-time data pipelines and large-scale streaming system design
- ▶ Introduction to Cybersecurity — secure system and sandboxed environment design principles
- ▶ OpenHack 2026 — Certificate of Participation, AI/ML Track
- ▶ Codeathon — Certificate of Participation, AiFi · AIML Club, SoCIT

Available 20+ hrs/week · Open to immediate start · Code samples & project repos available on request